USER INTERFACE DESIGN

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EXPERIMENT 2

Design a UI where users recall visual elements (e.g., icons). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION FRAME:

Chunking Analysis of the Instruction Page :

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:

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1. ***Clear and Sequential Numbering :***
   - The instructions are broken down into six steps, making it easier to follow.
   - Each step presents one key action in a structured manner, reducing cognitive overload.

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2. ***Logical Grouping of Information :***
   - **Observation Phase (Steps 1-2)** → Users learn about what they will see.
   - **Memorization Strategy (Step 3)** → Encourages users to focus on remembering items.
   - **Recall Phase (Steps 4-5)** → Explains how users will recall information.

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3. ***Visual Hierarchy and Design Elements :***

- **Bold, large title ("MEMORY RECALL TASK")** → Grabs attention and clearly states the task.
- **Bullet points and spacing** → Reduce clutter, enhancing readability.
- **Highlighted "START" button** → Signals the next step, keeping navigation intuitive.
- **Space-themed visuals** → Add a playful touch, engaging users without distraction.

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4. ***Time Constraint Reinforcement :***

- Step 5 explicitly states, "**You will have 5 seconds to view the items.**"
- This reinforces expectations while subtly urging users to focus.
- A Progressive bar also decreases for each second ensuring user to focus on time constraint.

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5. ***Simplicity and Clarity :***

- Instructions use short, direct sentences, avoiding unnecessary complexity.
- The active voice makes it more engaging and action-oriented.
- More clarity helps users to remember the icons

FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen :

This screen represents the Chunking Phase of a Memory Recall Task, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

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1. Purpose of the Screen

- This is the visual memory encoding phase, where users view and group items mentally before recalling them.

- The term "Chunking Phase" suggests that the game applies the chunking principle, which helps users remember items by categorizing them into meaningful groups.

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2. Key Elements and UI Components :

- ⏳ ***Countdown Timer (Top Left - Purple Circle: "00:05") :***

  - Indicates that users have 5 seconds to observe and memorize the displayed items.
  - The bright purple colour and bold text create urgency, ensuring users stay focused.

- ▯ ***Progress Bar (White Bar Below Title) :***

  - Shows the time remaining visually, reinforcing the sense of urgency
  - A partially filled bar suggests that some time has already passed.

- ☐ ***Grid of Icons :***

  - Various icons (such as a stars , astronaut , comet , galaxies , rocket, thunder etc.) are displayed in a 5x4 grid format.
  - These icons are visually distinct yet grouped by similarities, encouraging chunking strategies
  - Categorizing by theme (e.g. space , science and galaxies).
  - Grouping similar colours or backgrounds (e.g., purple and white tiles).
  - Associating repeated icons (e.g., multiple rockets, stars, comet).

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3. How the Chunking Phase Works :

- Users scan the grid and look for patterns or related items to create mental chunks.
- The countdown timer limits observation time, forcing quick memory strategies.
- Once time is up, users transition to the recall phase, where they must identify previously seen items

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4. Cognitive and UX Benefits of Chunking :

- Enhances short-term memory by allowing users to recall groups of information instead of individual elements.
- Reduces cognitive overload by helping users organize data efficiently.
- Improves pattern recognition, making recall easier and more accurate.

FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase :

This screen represents the Selection Phase of a Memory Recall Task, where users recall and choose the items they remember from the previous Chunking Phase. Below is a breakdown of its components:

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1. Purpose of the Screen :

- This is the memory retrieval stage, where users select the items they remember seeing in the previous phase.

- The goal is to test the effectiveness of chunking and short-term memory retention

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2. Key Elements and UI Components :

➢ 📝 ***Title & Instructions :***

- "**MEMORY RECALL TASK**" (Bold Header) – Reinforces the purpose of the task.

- "**SELECT THE ITEMS YOU REMEMBER:**" – Clear instruction guiding the user to choose remembered items.

➢ ☐ ***Grid of Icon Choices :***

- A set of eight icons are presented as multiple-choice options.

- Some icons were previously displayed, while others are distractors (new icons added to confuse users).

- Users must identify which icons appeared in the Chunking Phase.

- ✓ Check Box :

  - A component set containing checkboxes of empty and filled checkboxes are created

  - The component set is renamed as TRUE and FALSE respectively

  - The asset is added in frame

- ***"SUBMIT" Button (Bottom Center) :***

  - Once users have made their selections, they press "SUBMIT" to confirm their recall choices.

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3. How the Selection Phase Works :

- Users analyze the displayed icons and recall which ones they saw in the Chunking Phase.

- They select the remembered icons using the radio buttons below each option.

- Some icons are distractors, testing whether the user's memory is accurate or if they mistakenly recall incorrect icons.

- Clicking "SUBMIT" finalizes their choices, leading to a results or feedback screen.

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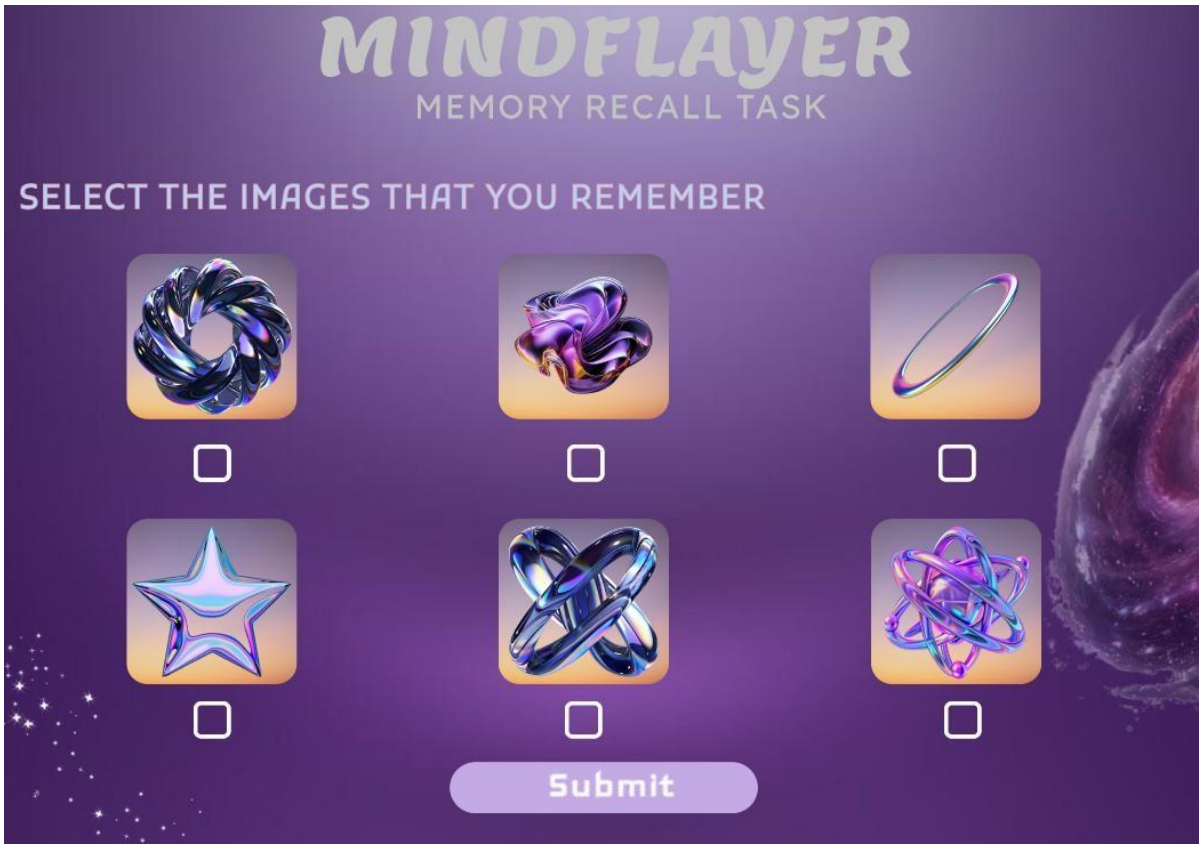
4. Cognitive and UX Benefits :

- Tests memory accuracy by comparing user selections with previously displayed items.

- Incorporates distractors to evaluate how well users distinguish real vs. false memories.



FRAME 4:

REMEMBER PHASE:

Analysis of the Memory Recall Task - Score & Feedback Screen :

This screen represents the Score s Feedback Phase of the Memory Recall Task, where users receive their performance evaluation based on the selections made in the previous Recall Phase.

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1. ***Purpose of the Screen :***

- Provides feedback on recall accuracy by showing the number of correct answers.
- Allows users to decide their next action (continue, restart, or exit).

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2. ***Key Elements & UI Components :***

- 📝 ***Title & Score Display :***

  - "MEMORY RECALL TASK" – Reinforces the game's title.
  - "YOUR RECALL SCORE!" – Indicates that the user's performance is being displayed.

- ***Action Buttons (Right Side) :***

  - "RESTART" – Allows the user to retake the memory recall test.
  - "EXIT" – End Task and Exits Game.

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3. ***How This Phase Works :***

- A variable (number) is created and assigned to score value
- Another prototype is added to checkbox present in "RECALL FRAME"
- The prototype action is increased or decreased as per image in both frames
- It calculates the accuracy score and displays it on a digital number
- Users review their performance and choose their next action either Continue or Exit

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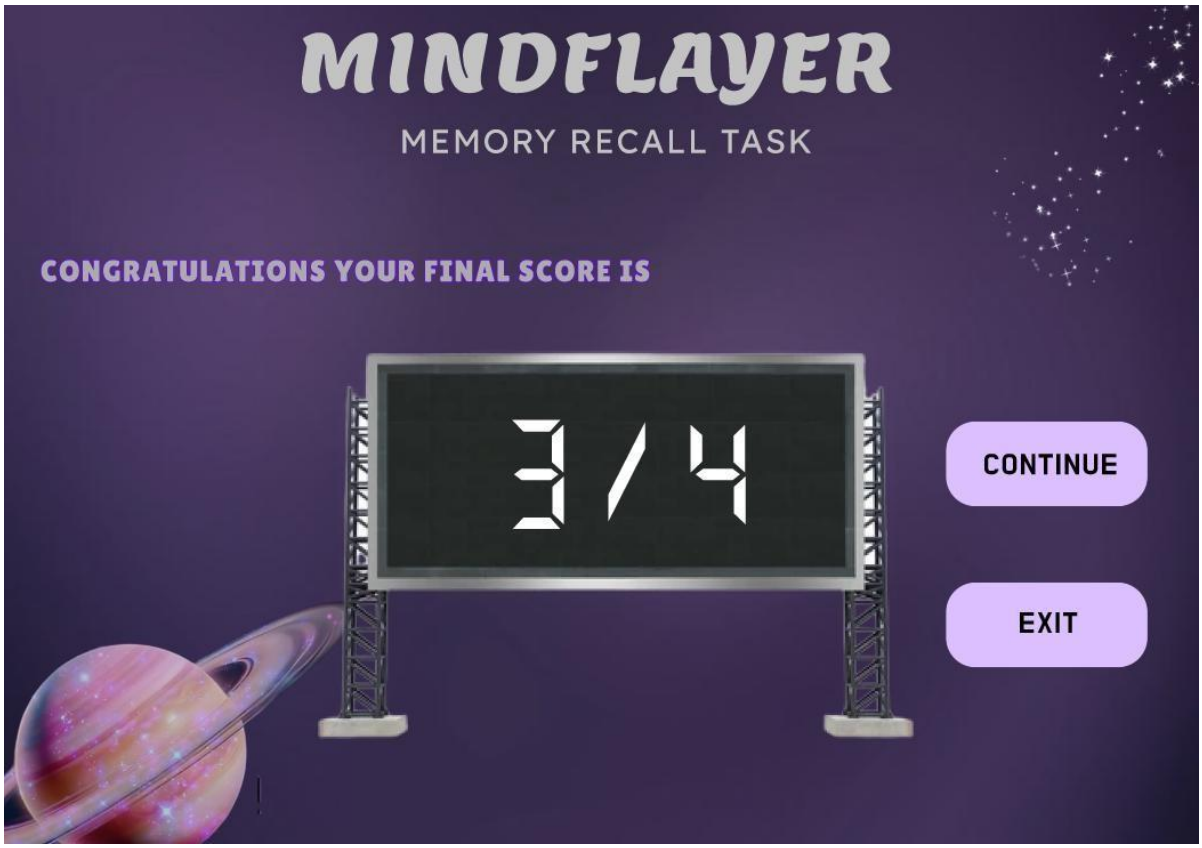
4. ***Cognitive & UX Benefits :***

- Instant feedback helps users track their memory performance.
- Multiple options (Continue, Exit) give users control over their learning experience.



PROTOTYPE LINK:

https://www.figma.com/proto/geStc55wc5CMlVQ1Jg9Jdb/EXERCISE-1-CUP-CAKES?node-id=0-1&t=w9bOmT5LzSgmAkjN-1